

Course Code	21CSC305P	Course Name	MACHINE LEARNING	Course Category	C	PROFESSIONAL CORE	L	T	P	C
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Pre-requisite Courses	Nil	Co- requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	School of Computing	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):		The purpose of learning this course is to:		Program Outcomes (PO)												Program Specific Outcomes		
CLR-1:				1	2	3	4	5	6	7	8	9	10	11	12			
CLR-1:	explore the fundamental mathematical concepts of machine learning algorithms			Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CLR-2:	apply linear machine learning model to perform regression and classification			-	2	-	-	-	-	-	-	-	-	-	-	1	-	-
CLR-3:	utilize mixture models to group similar data items			-	2	-	2	-	-	-	-	-	-	-	-	-	-	2
CLR-4:	develop machine learning models for time –series data prediction			-	3	-	3	-	-	-	-	-	-	-	-	-	-	2
CLR-5:	design ensemble learning models using various machine learning algorithms			-	3	-	3	-	-	-	-	-	-	-	-	-	-	3
CO-1:	understand the basics of machine learning using probability theory			-	3		3	-	-	-	-	-	-	-	-	-	-	3
CO-2:	implement machine learning models using supervised learning algorithms			-	3		3	-	-	-	-	-	-	-	-	-	-	3
CO-3:	implement machine learning models using unsupervised learning algorithms			-	3		3	-	-	-	-	-	-	-	-	-	-	3
CO-4:	implement machine learning models for sequential data analysis and prediction			-	3		3	-	-	-	-	-	-	-	-	-	-	3
CO-5:	develop ensemble learning models for supervised and unsupervised learning			-	3		3	-	-	-	-	-	-	-	-	-	-	3

Unit-1 - Introduction	9 Hour
machine learning what and why?, supervised and unsupervised learning, polynomial curve fitting, probability theory- discrete random variables, fundamental rules,Bayes rule, Independence and conditional independence, continuous random variables, Quantiles, Mean and variance, probability densities, Expectation and covariance.	
Practice:	
1. Devise a program to import, load and view dataset	
2. Create a program to display the summary and statistics of the dataset	
Unit-2 - Linear models for Regression	9 Hour
Maximum likelihood estimation – least squares, robust linear expression, ridge regression, Bayesian linear regression. Linear models for classification: Discriminant function – Probabilistic generative models, Probabilistic discriminative models, Laplacian approximation, Bayesian logistic regression, Kernels functions, using kernels in GLMs, Kernel trick, SVMs.	
Practice:	
1. Implement linear regression to perform prediction	
2. Implement Bayesian logistic regression and SVM for classification	
Unit-3 - Mixture Models and EM	9 Hour
K-means clustering, mixtures of Gaussians, An alternative view of EM, Factor analysis, PCA, choosing the number of latent dimensions. Clustering – measuring dissimilarity, evaluating the output of clustering methods, Hierarchical clustering.	
Practice:	
1. Implement K-means clustering, mixtures of Gaussians and Hierarchical clustering algorithm to categorize data.	
2. Create a program to perform PCA	

Unit-4 – Hidden Markov Models **9 Hour**

Sequential data – Markov models, HMM – maximum likelihood for the HMM, The forward and Backward algorithm, the sum-product algorithm, scaling factors, Viterbi algorithm, linear dynamical systems.

Practice:

1. Implement HMM to predict the sequential data

Unit-5 – Combining Models **9 Hour**

Bayesian model averaging, Boosting, Adaptive basis function models, CART, generalized additive models, Ensemble learning.

Practice:

1. Implement CART learning algorithms to perform categorization
2. Implement Ensemble learning models to perform classification

Learning Resources	1. Pattern Recognition and Machine Learning, Christopher M Bishop, Springer, 2006.	2. Machine Learning- A probabilistic perspective, Kevin P.Murphy, The MIT Press, 2012.
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Learning Assessment									
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)						Final Examination (0% weightage)	
		Formative CLA-1 Average of unit test (20%)		Project Based Learning CLA-2 (60%)		Report and Viva Voce (20% weightage)			
		Theory	Practice	Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	15%	-	-	15%	-	15%	-	-
Level 2	Understand	25%	-	-	20%	-	20%	-	-
Level 3	Apply	30%	-	-	25%	-	25%	-	-
Level 4	Analyze	30%	-	-	25%	-	25%	-	-
Level 5	Evaluate	-	-	-	10%	-	10%	-	-
Level 6	Create	-	-	-	5%	-	5%	-	-
	Total	100 %		100 %		100 %		-	

Course Designers			
Experts from Industry		Experts from Higher Technical Institutions	Internal Experts
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